SCOREBOARD – Hadamard 14304 Paley-I (2026-10-09)

Cutoff: 2026-10-09T11:35:14Z (~7:35AM ET)

Object	Status	max|HHˆT-nI|	Wall (s)	SPK
H_14304 Paley-I (q=14303)	PASS	0	2725.7243	**IN SPK** (Paley-I)
H_14304 Paley-II (q=7151)	FAIL/SKIP	–	–	q=7151 prime but ===3 mod 4 needs ===1
H_14304 Kronecker (4*3576)	FAIL/SKIP	–	–	3576===0 mod 4 order-admissible; H_3576 has NO direct classical Paley route

Units: 678/678 * row blocks 112/112 * overall_pass=True * wall_total=2767.5075s
Gap 2052: **OPEN** * clay_closed=false * GATE-04 **LOCKED** * conjecture **OPEN**
IN SPK via Paley-I; Miyamoto digression skipped; exclude Miyamoto-only 8948/8644/8516/...; prior SPK 10224+10212+10164+10160+10152+10140+10112+10104+10100+10068+10040+10008+9968+9932+9924+9908+9888+9884 retained

Next optional: **q2=6961->n=13924 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=6961 prime ===1 mod 4; Paley-I q=13923=3ˆ2*71317 NOT prime -> FAIL; Kronecker 13924=4*3481; 3481=59ˆ2 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3481); 13924=2ˆ2*59ˆ2 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=6977->n=13956 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=6977 prime ===1 mod 4; Paley-I q=13955=5*2791 NOT prime -> FAIL; Kronecker 13956=4*3489; 3489=3*1163 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3489); 13956=2ˆ2*31163 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=6997->n=13996 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=6997 prime ===1 mod 4; Paley-I q=13995=3ˆ2*5*311 NOT prime -> FAIL; Kronecker 13996=4*3499; 3499 odd prime (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3499); 13996=2ˆ2*3499 ===12 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=7001->n=14004 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7001 prime ===1 mod 4; Paley-I q=14003=11*1967 NOT prime -> FAIL; Kronecker 14004=4*3501; 3501=3*2389 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3501); 14004=2ˆ2*3*2389 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=7013->n=14028 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7013 prime ===1 mod 4; Paley-I q=14027=13ˆ2*83 NOT prime -> FAIL; Kronecker 14028=4*3507; 3507=3*7167 odd (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3507); 14028=2ˆ2*37167 ===12 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=7057->n=14116 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7057 prime ===1 mod 4; Paley-I q=14115=3*5941 NOT prime -> FAIL; Kronecker 14116=4*3529; 3529 odd prime (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3529); 14116=2ˆ2*3529 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=7069->n=14140 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7069 prime ===1 mod 4; Paley-I q=14139=3ˆ2*1571 NOT prime -> FAIL; Kronecker 14140=4*3535; 3535=5*7101 odd (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3535); 14140=2ˆ2*5*7101 ===12 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=7109->n=14220 Paley-II DIRECT (skipped by the Paley-I-only chain;

q2=7109 prime ===1 mod 4; Paley-I q=14219=59*241 NOT prime -> FAIL; Kronecker 14220=4*3555; 3555=3^2*5*79 odd (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3555); 14220=2^2*3*5*79 ===12 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=7129->n=14260 Paley-II DIRECT (skipped by t
NOT prime -> FAIL; Kronecker 14260=4*3565; 3565=5*23*31 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3565); 14260=2^2*5*23*31 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | next Paley-I DIRECT: q=14323->n=14324 (prime ===3 mod 4; no prime ===3 mod 4 in 14304..14322; 14321 ===1 skipped; Paley-II q2=7161=3*7*11*31 NOT prime power (===1 mod 4 but composite) -> FAIL/SKIP; Kronecker 14324=4*3581; 3581 odd prime (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3581); 14324=2^2*3581 ===4 mod 16; FIRST any direct route after 14304) | Kronecker-only (no direct Paley route for n; note-only, NOT on box): 14320=4*3580 (H_3580 via Paley-II q2=1789 prime ===1 mod 4; 14320=2^5*179; Paley-I q=14319=3^2*37*43 NOT prime; Paley-II q2=7159 prime but ===3 mod 4) – 14308=2^2*7^2*73, 14312=2^3*1789 (===8 mod 16), 14316=2^2*3*1193 have NO direct route and m=n/4 not Hadamard-admissible -> FAIL/SKIP | FIRST Paley-II DIRECT after 14304: q2=7177->n=14356 (q2=7177 prime ===1 mod 4; Paley-I q=14355=3^2*5*11*29 NOT prime -> FAIL; Kronecker 14356=4*3589; 3589=37*97 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3589); 14356=2^2*37*97 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII); comes AFTER the next Paley-I 14324) (or gap 2052 / Track B)**